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# Emerging trends and global collaboration in Paclitaxel resistance research for breast cancer: a comprehensive bibliometric study

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## Abstract

Paclitaxel, a compound that targets microtubules in cancer cells and inhibits mitosis, has been used as a first-line treatment for recurrent or metastatic breast cancer for over three decades. Research into paclitaxel resistance has yielded thousands of articles. However, this body of research has not been analyzed systematically and quantitatively. This study employs bibliometric analysis and visualisation techniques using VOSviewer and CiteSpace software to elucidate the contributions from countries and institutions, collaboration patterns, journal publication trends, research evolution, and current hotspots. Our findings reveal that leading research institutions in the United States, Europe, and East Asia hold significant advantages in this field. Specifically, the United States contributed the most to this field, with MD Anderson Cancer Center being the most prominent research institution and having a broad collaboration network. *Prestigious oncology* journals serve as primary platforms for disseminating key findings, with *Cancer Research* publishing the most relevant articles and having the highest co-citation frequency. We identified the main research branches by analysing keyword co-occurrence and keyword bursts, including treatment methods, drug delivery, mechanisms, and cellular response. Current research focuses on unveiling resistance mechanisms and exploring new therapeutic strategies. Paclitaxel resistance in breast cancer, particularly in TNBC and metastatic cases, remains an active and critical research area. Integrating nanotechnology into drug development and therapeutic strategies is a current focal point and promises to be a significant future direction. These insights highlight a strategic direction for future research, including strengthening international partnerships, deepening understanding of resistance mechanisms, and applying engineered therapeutics for targeted breast cancer treatments.

**Keywords** Paclitaxel, Resistance, Breast cancer, Bibliometrics, Nanotechnology

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## 1 Introduction

Breast cancer stands as the most prevalent malignant tumor among women globally, posing a significant threat to their health. In 2022, there were approximately 2.3 million new cases diagnosed and 660,000 deaths [1], underscoring its status as a leading cause of mortality. The main treatment options include surgery, chemotherapy, radiotherapy, and molecular targeted therapy, each playing a critical role in managing the disease.

Paclitaxel is a natural diterpenoid compound isolated from the yew tree [2]. It acts as an antimitotic agent by selectively binding to tubulin, thereby regulating the expression of cyclins and apoptosis-related proteins [3], which ultimately induces cell apoptosis. Since its approval by the FDA in 1993 [4], paclitaxel has been extensively used in the treatment of various cancers, including breast, ovarian, lung, and head and neck cancers, owing to its potent antitumor effects. It is currently regarded as a first-line chemotherapy agent for advanced and metastatic breast cancer.

The application of paclitaxel in breast cancer is commonly tailored according to the disease stage and molecular subtype of patients. For early-stage breast cancer, paclitaxel is often combined with anthracyclines to reduce the risk of distant metastasis [5]. In cases of locally advanced breast cancer, it is used in combination with anthracyclines or alkylating agents as part of neoadjuvant chemotherapy to shrink the primary tumor and regional lymph nodes, enabling previously inoperable patients to undergo surgery [6]. In advanced or metastatic breast cancer, paclitaxel is an essential option for first-line or second-line salvage therapy, either as monotherapy or in combination with targeted drugs, to control tumor progression and prolong survival [7]. In terms of molecular subgroups, paclitaxel demonstrates significant therapeutic value across different categories of breast cancer. Triple-negative breast cancer (TNBC), which lacks expression of ER, PR, and HER2, has limited options for targeted therapy. Paclitaxel is a cornerstone in adjuvant, neoadjuvant, and metastatic treatment regimens for TNBC [8]. For HER2-positive breast cancer, paclitaxel is frequently combined with anti-HER2 agents, such as trastuzumab and pertuzumab, significantly enhancing efficacy in both early adjuvant and late salvage settings [9]. Additionally, in hormone receptor-positive (HR+)/HER2-negative cases where endocrine resistance or rapid tumor progression is observed, paclitaxel may be incorporated into the chemotherapy regimen [10].

However, prolonged administration often leads to the development of resistance, reducing therapeutic efficacy [11]. The definition and diagnosis of paclitaxel resistance are typically based on clinical evaluation and biomarker analysis. Clinically, resistance is defined by tumor progression or recurrence following an initial positive response to treatment [12]. At the molecular level, several mechanisms have been implicated in paclitaxel resistance, such as the overexpression of TUBB3, which interferes with drug binding [13]; increased expression of P-glycoprotein, which enhances drug efflux [14]; activation of the PI3K/Akt signaling pathway and aberrant expression of epithelial-mesenchymal transition markers, which may reflect the involvement of specific signaling pathways in drug resistance [15].

Over the past three decades, extensive research on paclitaxel resistance in breast cancer has been carried out, resulting in thousands of publications covering various aspects such as drug resistance profiles [16], mechanisms of resistance [17], and clinical trial outcomes [18]. However, this body of research has not been systematically and quantitatively analyzed, particularly regarding contributions from research institutions, the excellent publication platforms, and the evolution of research topics and emerging hotspots. A systematic overview of the spatiotemporal research landscape in this field can provide valuable references for researchers to identify suitable collaborators, select appropriate journals for publication, and promptly grasp current research frontiers and development trends.

Bibliometric analysis, a rapidly evolving discipline, provides an effective means of addressing this challenge. By utilizing text mining techniques on scholarly literature, it facilitates the extraction of valuable information and enables quantitative and statistical analysis of publications within specific research domains. Furthermore, it supports the visual representation of research interests [19]. This study employed bibliometric analysis to quantitatively examine contributions from various countries and institutions, collaboration networks, publication trends across journals, the evolution of research themes, and current research hotspots. Additionally, this study integrates existing findings with potential future research directions to offer a temporal-geography landscape of the field's development.

## 2 Materials and methods

### 2.1 Data collection

Due to its higher authority and suitability for bibliometric analysis [20, 21], the Science Citation Index-Expanded (SCI-E) was selected as the data source of this study. The search formula used for indexing data was ((ALL=(paclitaxel)) AND ALL=(resistance)) AND ALL= (breast cancer), with a publication date range from January 1, 1993, to December 31, 2024. Data collection was completed on February 14, 2025. A total of 3687 documents were retrieved from the SCI-E. Since the primary focus of the study was on articles, reviews, and conference abstracts, other types of documents and duplicates were excluded. It resulted in a final dataset of 3,668 documents in citation format (.txt files) for further analysis (Supplementary Fig.S1).

### 2.2 Data analysis

The study employed Microsoft Office Excel to analyze the distribution of literature types, annual publication output, and the overall status of journals publishing relevant articles. VOSviewer was chosen for country/institution collaboration analysis and keyword clustering, owing to its robust capabilities in visualizing co-occurrence networks and generating precise clusters. By fine-tuning parameters, nodes with minimal influence on the network structure were filtered out, emphasizing the network's core relationships and improving interpretability. Specifically, the country collaboration network included only countries with five or more publications, while the institutional collaboration network focused on institutions with ten or more publications. For keyword clustering, only keywords appearing at least 20 times were included, with a clustering resolution set at 1.0. CiteSpace was utilized to visualize highly cited journals and detect emerging citation trends, capitalizing on its strengths in time-series analysis and identifying emerging research themes. The parameter settings included a time slicing interval of one year, a selection threshold of the top 50 nodes per slice, and a g-index scale factor k of 25 (Supplementary Figure S1).

To ensure the reliability of the results, two validation methods were implemented: (1) a random sample of 5% of the literature (183 articles) was manually reviewed for consistency in keyword extraction and clustering, yielding a consistency rate of 91%; and (2) by adjusting the clustering resolution in VOSviewer between 0.7 and 1.0, it was confirmed that the core cluster structure remained stable, demonstrating the robustness of the analytical outcomes [22].

## 3 Results

### 3.1 The general situation of the publication

A total of 3668 articles related to paclitaxel resistance in breast cancer were retrieved from the SCI-E database, comprising 3132 research articles (85.4%), 489 review articles (13.3%), and 47 conference abstracts (1.3%) (Fig. 1a). Annual publication output showed sustained growth over past 31 years, increasing from 4 to approximately 220 publications (Fig. 1b). From 1993 to 2007, when clinical use of paclitaxel was still limited and resistance has not yet become a widespread issue, the number of annual publications remained below 100. Rising breast cancer incidence and increasing prevalence of resistance later stimulated greater research interest, leading to peak growth by 2017. However, during the most recent eight-year period (2017–2024), publication volume stabilized at between 200 and 250 papers annually (Fig. 1b). This plateau may indicate

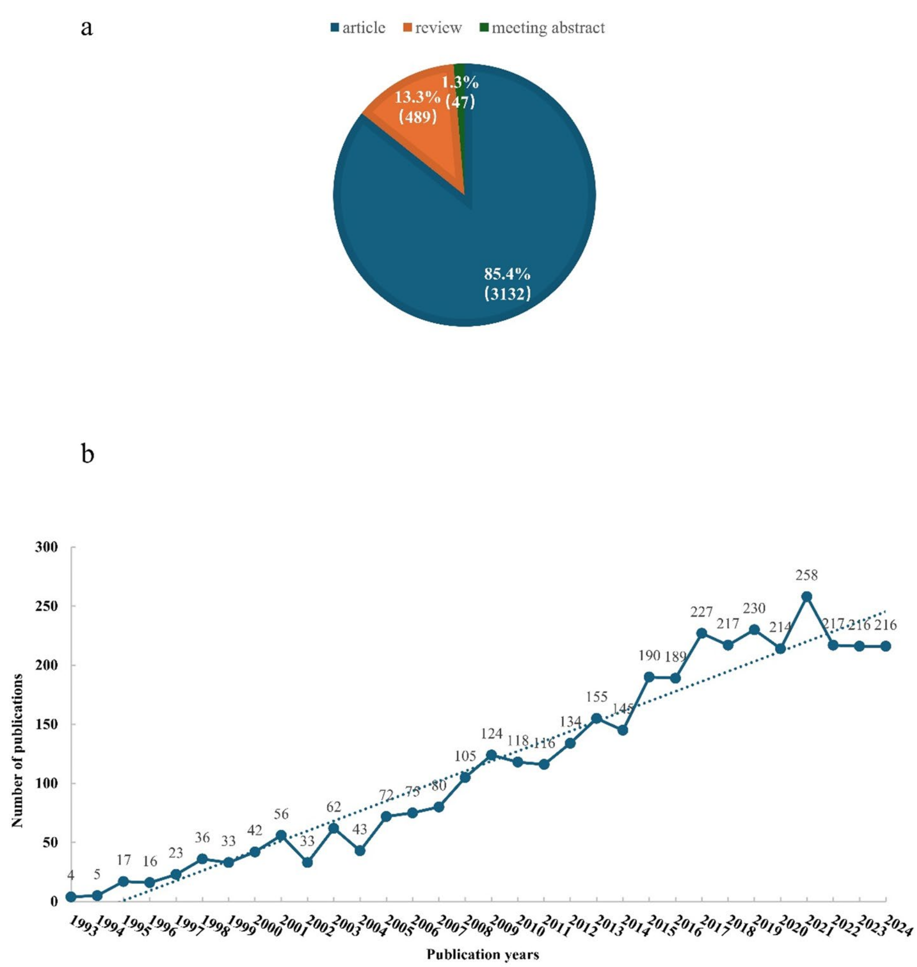


**Fig. 1** The general situation of the publication. **a** The composition of document types. (Documents were retrieved from the Science Citation Index-Expanded (SCI-E) covering publications from January 1, 1993, to December 31, 2024. The data collection process was finalized on February 14, 2025). **b** Annual number of publications

persistent challenges in overcoming the mechanistic barriers underlying paclitaxel resistance, as well as a shift in research focus toward alternative therapeutic strategies, such as targeted therapy and immunotherapy [23]. Nonetheless, the consistent annual output of more than 200 publications confirms that paclitaxel resistance in breast cancer remains a significant and actively investigated research domain.

### 3.2 Contributions from countries/regions and institutions

#### *3.2.1 Contributions from countries/regions*

To construct a comprehensive knowledge map of countries and institutions researching paclitaxel resistance in breast cancer, we analyzed the affiliations of 3668 articles originating from 89 countries/regions. The United States leads with 1,237 publications (33.72%), followed by China with 1,175 publications (32.03%). Italy, Japan, Canada, England, South Korea, India, France, and Germany rank third to tenth, respectively, contributing 175 (4.77%), 166 (4.53%), 157 (4.28%), 151 (4.12%), 148 (4.03%), 145 (3.95%), 140 (3.82%), and 135 (3.68%) publications (Table 1).

The number of citations is another critical metric for evaluating the contributions and impact within a specific research domain [24]. According to Table 1, the United States leads with a total citation count of 93,801, followed by China with 40,283 citations. Canada, England, Italy, Japan, Germany, France, the Netherlands, and Spain rank third through tenth in citation counts (Table 1). Considering the variations in publication output across different countries, we also assessed the average number of citations per paper. Notably, despite producing only 56 publications, the Netherlands achieved an impressive mean of 120 citations, the highest among the analyzed countries (Table 1). Similarly, Singapore recorded the second-highest average citation counts of 99, based on just 29 publications (data not shown), indicating the high quality of its research output (Table 1). The United States averages 76 citations per paper, slightly lower than England

**Table 1** The top 20 countries/regions ranked by publications and citations

| Rank | Publication | | | Citation | | |
|---|---|---|---|---|---|---|
| | Country | Count | Percentage | Country | Count | Average citations |
| 1 | USA | 1237 | 33.72% | USA | 93,801 | 76 |
| 2 | CHINA | 1175 | 32.03% | CHINA | 40,283 | 34 |
| 3 | ITALY | 175 | 4.77% | CANADA | 12,213 | 78 |
| 4 | JAPAN | 166 | 4.53% | ENGLAND | 11,858 | 79 |
| 5 | CANADA | 157 | 4.28% | ITALY | 9881 | 56 |
| 6 | ENGLAND | 151 | 4.12% | JAPAN | 7860 | 47 |
| 7 | SOUTH KOREA | 148 | 4.03% | GERMANY | 7492 | 55 |
| 8 | INDIA | 145 | 3.95% | FRANCE | 7426 | 53 |
| 9 | FRANCE | 140 | 3.82% | NETHERLANDS | 6698 | 120 |
| 10 | GERMANY | 135 | 3.68% | SPAIN | 5890 | 62 |
| 11 | SPAIN | 95 | 2.59% | SOUTH KOREA | 4973 | 34 |
| 12 | CHINA TAIWAN | 80 | 2.18% | AUSTRALIA | 4294 | 60 |
| 13 | AUSTRALIA | 71 | 1.94% | INDIA | 4023 | 28 |
| 14 | IRAN | 64 | 1.74% | SWITZERLAND | 3329 | 58 |
| 15 | SWITZERLAND | 57 | 1.55% | CHINA TAIWAN | 3111 | 39 |
| 16 | NETHERLANDS | 56 | 1.53% | BELGIUM | 2988 | 65 |
| 17 | TURKEY | 54 | 1.47% | SINGAPORE | 2866 | 99 |
| 18 | POLAND | 51 | 1.39% | IRAN | 2228 | 35 |
| 19 | BRAZIL | 48 | 1.31% | TURKEY | 2024 | 37 |
| 20 | BELGIUM | 46 | 1.25% | POLAND | 1971 | 39 |

(79) and Canada (78), which rank third and fourth, respectively. However, China's average citation rate was 34, marginally higher than India's (28) among the top twenty countries (Table 1).

These statistics reveal that the top 20 countries and regions are predominantly from Europe, North America, and East Asia. The distribution of resources and talents in this field has obvious geographical agglomeration, which is highly consistent with the core areas of global cancer research. The United States is the core leader in this field. China was a significant producer, but its citation impact was still relatively limited, suggesting that there was still room for improvement in its research's innovation or translational value. Some small countries have achieved high impact by focusing on high-value issues, providing multiple paths for development in the field.

#### *3.2.2 Contributions from institutions*

A total of 4,059 institutions globally have contributed to research on paclitaxel resistance in breast cancer. As illustrated in Table 2, the top 20 institutions by publication volume are primarily located in China and the United States. Specifically, 11 institutions are from China, and the remaining nine are from the United States. Notably, MD Anderson Cancer Center leads with 74 publications. Fudan University follows closely with 68 publications, and Zhejiang University ranks third with 63 papers (Table 2). The other top ten institutions for publication counts are Sun Yat-sen University, National Cancer Institute, Chinese Academy of Sciences, Nanjing Medical University, The University of Texas, Shanghai Jiao Tong University, and Memorial Sloan Kettering Cancer Center. In China, all leading contributors are universities, whereas in the United States, they mainly comprise universities and research centers (Table 2).

Regarding citations, the Dana-Farber Cancer Institute leads with 8,629 citations, an average of 215.73 citations per publication. The University of Texas follows closely with 8,014 citations from 49 articles, while Harvard University ranks third with 41 articles, cumulating 7,350 citations. The other top ten institutions for citation counts are the National Cancer Institute, MD Anderson Cancer Center, Weill Cornell Medical College, Mayo Clinic, Houston Methodist Research Institute, Memorial Sloan Kettering Cancer Center, and Emory University (Table 2). The average number of citations per paper reflects the average impact of individual papers. It should be noted that Houston Methodist Research Institute ranked first with an average of 875 citations. Weill Cornell Medical College was cited 502.91 times on average, ranking second(Table 2). Although the total citation frequency was not among the top, the average citation frequency was far higher than other institutions. This indicates that the quality of its single article is extremely high. Their research results accurately touched on the key issues in the field and triggered in-depth discussion.

Only three of the top twenty most-cited institutions are from China: Sun Yat-sen University at 14th place, Fudan University at 16th place, and Zhejiang University at 17th place. Canada and Singapore contributed two institutions to this list, ranking 15th and 18th, respectively. The remaining institutions all come from the United States (Table 2). While Chinese institutions demonstrate strength in research output scale, their academic influence still lags significantly behind that of their American counterparts. In contrast, U.S. institutions excel not only in scale but also in research quality. This disparity fundamentally reflects differences in institutional research models: leading U.S.

**Table 2** The top 20 institutions ranked by publications and citations

| Rank | Publication | | | Citation | | | |
|---|---|---|---|---|---|---|---|
| | Institution | Count | Country | Institution | Count | Average citations | Country |
| 1 | MD Anderson Cancer Center | 74 | USA | Dana-Farber Cancer Institute | 8629 | 215.73 | USA |
| 2 | Fudan University | 68 | CHINA | The University of Texas | 8014 | 163.55 | USA |
| 3 | Zhejiang University | 63 | CHINA | Harvard University | 7350 | 179.27 | USA |
| 4 | Sun Yat-sen University | 62 | CHINA | National Cancer Institute | 6376 | 111.86 | USA |
| 5 | National Cancer Institute | 57 | USA | MD Anderson Cancer Center | 5626 | 76.03 | USA |
| 6 | Chinese Academy of Sciences | 53 | CHINA | Weill Cornell Medical College | 5532 | 502.91 | USA |
| 7 | Nanjing Medical University | 51 | CHINA | Mayo Clinic | 5358 | 198.44 | USA |
| 8 | The University of Texas | 49 | USA | Houston Methodist Research Institute | 5250 | 875 | USA |
| 9 | Shanghai Jiao Tong University | 47 | CHINA | Memorial Sloan Kettering Cancer Center | 5172 | 114.93 | USA |
| 10 | Memorial Sloan Kettering Cancer Center | 45 | USA | Emory University | 4934 | 214.52 | USA |
| 11 | China Medical University | 44 | CHINA | Massachusetts General Hospital | 4669 | 113.88 | USA |
| 12 | Sichuan University | 42 | CHINA | Northwestern University | 4203 | 200.14 | USA |
| 13 | Harvard University | 41 | USA | Indiana University | 3583 | 275.66 | USA |
| 14 | Massachusetts General Hospital | 41 | USA | Sun Yat-sen University | 3436 | 55.42 | CHINA |
| 15 | Shandong University | 41 | CHINA | British Columbia Cancer Agency | 3326 | 475.14 | CANADA |
| 16 | Dana-Farber Cancer Institute | 40 | USA | Fudan University | 2927 | 43.04 | CHINA |
| 17 | Chinese Academy of Medical Sciences | 37 | CHINA | Zhejiang University | 2706 | 42.95 | CHINA |
| 18 | University of Pittsburgh | 36 | USA | National University of Singapore | 2575 | 151.47 | SINGAPORE |
| 19 | Zhengzhou University | 34 | CHINA | Fox Chase Cancer Center | 2503 | 139.06 | USA |
| 20 | Harvard Medical School | 30 | USA | University of North Carolina | 2472 | 107.48 | USA |

institutions tend to focus more on core scientific challenges and exhibit higher efficiency in translating research into impact, whereas Chinese institutions remain in a stage of "quantitative accumulation" and need to further enhance the originality of their research and their international collaborative capacity.

Furthermore, it is noteworthy that most top-ranked institutions regarding citation impact are research centers rather than comprehensive universities. Although these centers may publish fewer papers overall, the citation frequency per document tends to be higher. This phenomenon could be attributed to their more focused research directions and closer integration with clinical practice. It highlights the advantage of specialized institutions in achieving greater research depth within niche fields (Table 2).

## 3.3 Collaboration status of the countries/territories and institutions

### 3.3.1 The collaboration status of the countries/territories

Collaboration is pivotal in contemporary scientific research, fostering academic exchange and expediting advancements. A key objective in collaborative network analysis is the identification of the most influential nodes. Within such networks, nodes characterized by frequent collaboration and extensive partnerships typically demonstrate higher centrality scores, reflecting their prominent positions. Centrality analysis at the country level can effectively reveal their influence within this research domain [21, 25]. Using CiteSpace software, we calculated the centrality scores for 89 countries or territories.

The United States possesses the highest centrality score of 0.36, with 49 collaboration partners, highlighting its central position in this research field (Supplementary Table S1). Countries with scores from 2nd to 10th are England, Italy, Germany, Spain, France, Australia, Egypt, Switzerland, and Japan. At the same time, India, Netherlands, Saudi Arabia, Malaysia, China Taiwan Province, Belgium, Greece, China, Canada, and Poland have centrality scores ranking 11th to 20th, respectively (Supplementary Table S1). It is worth noting that although China, Canada, and Poland have many partners (41, 37, and 36, respectively), they possess relatively low centrality, only 0.03 (Supplementary Table S1). Although they engage in extensive collaboration, they do not serve as a core link for cross-group partnership, reflecting the breadth but lack of depth of collaboration (Supplementary Table S1).

We then analyzed the collaboration pattern among 55 countries that have published five or more articles using VOSviewer software. The co-occurrence network diagram reveals that these collaboration patterns can be classified into eight categories. Specifically, the United States, China, India, and Malaysia form one category; Germany, Italy, and Spain constitute another group; France, Russia, Ireland, Finland, and Greece create a third cluster. Japan is clustered with Egypt and Saudi Arabia. Meanwhile, South Korea is grouped with Iran, Turkey, Poland, Mexico, Portugal, Hungary, and Singapore. England, New Zealand, and Scotland are categorized together, while Thailand and Brazil each represent their distinct group (Fig. 2a).

Regarding the intensity of overall cooperation, European nations demonstrate tighter collaboration than other regions. The British Isles countries closely cooperate with European countries. It is conducive to knowledge sharing and collaborative innovation, which may improve the overall impact of research in the region (Fig. 2a). In Asia, while China, Japan, South Korea, and India are major research powers, their level of collaboration is generally less tight than among Germany, Italy, and France. However, China stands out in terms of a broader scope of cooperation (Fig. 2a). It not only collaborates extensively within Asia, including with Japan, India, South Korea, and Thailand, but also maintains strong partnerships with North American nations such as the United States and Canada, as well as European countries like England, Italy, and France (Fig. 2a).

### 3.3.2 Collaboration status of institutions

The centrality scores for 4,059 institutions were also calculated using CiteSpace software. Supplementary Table 2 presents the top ten institutions based on centrality. Six institutions are in the United States, three in China (ranked 7th, 9th, and 10th), and one in France (ranked third) (Supplementary Table S2). Notably, the top two institutions are

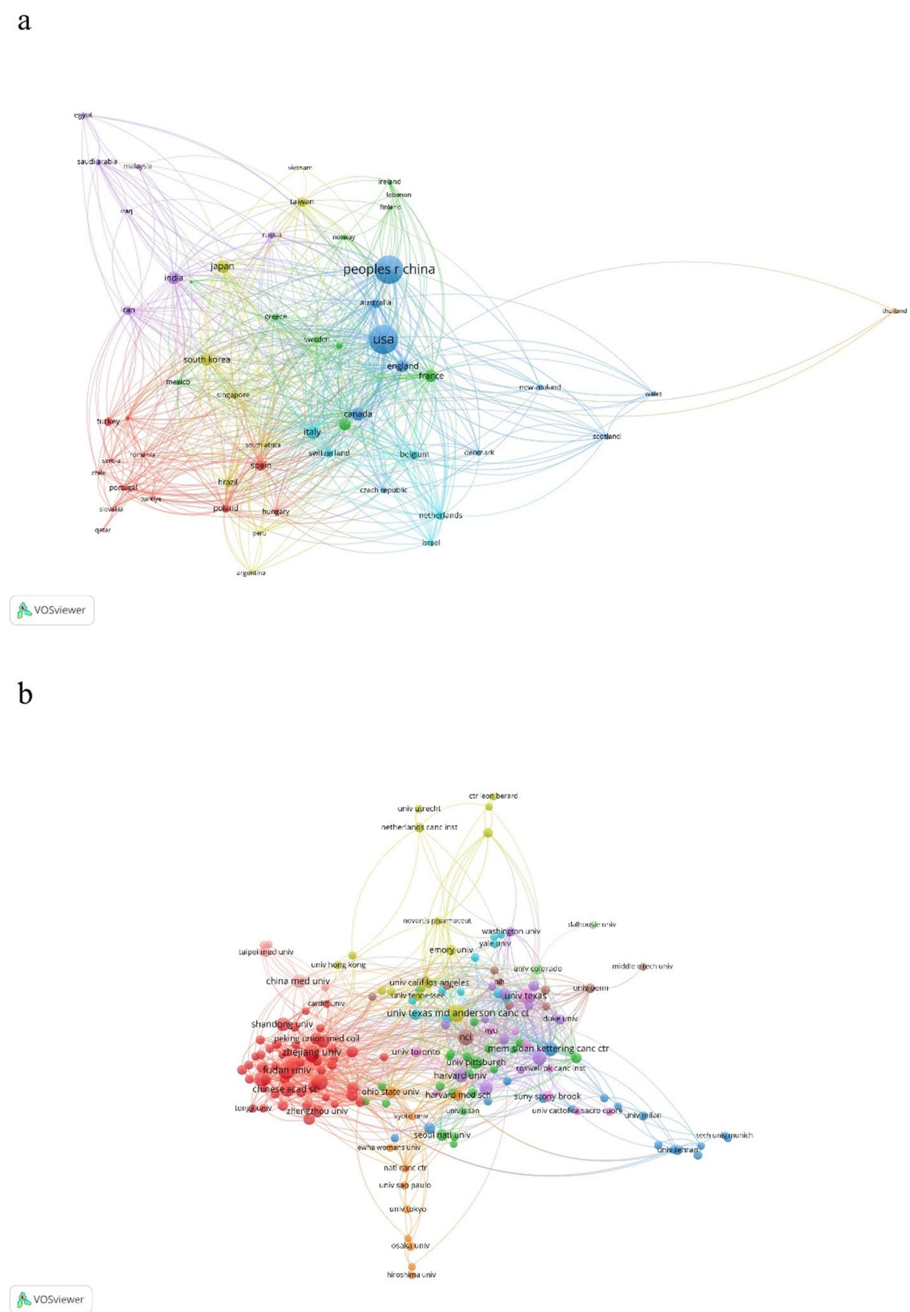


**Fig. 2** The co-occurrence network diagram of collaboration. **a** Countries/regions' co-occurrence network. (In this visualization, colors denote clusters, dot size indicates publication numbers, and connecting lines signify the strength of partnerships.) **b**) Institutions' co-occurrence network

both from the United States. MD Anderson Cancer Center leads with a centrality score of 0.18, followed by the Dana-Farber Cancer Institute at 0.14. France's Institute Gustave-Roussy holds the third place, China's Sun Yat-sen University ranks seventh, the Chinese Academy of Sciences ranks ninth, and Fudan University tenth. Despite having only ten partners (the fewest among the top 10 countries), France's Institute Gustave Roussy has a significantly higher centrality score of 0.09 compared to the Chinese Academy of Sciences and Fudan University, both at 0.05. This reflects that although the scope of cooperation of professional cancer research institutions is narrower, the quality and influence of academic connections in specific fields are more prominent (Supplementary Table S2).

The co-occurrence network diagram in Fig. 2b provides an intuitive overview of international collaboration among 172 institutions that have each published at least 10 articles. Two distinct clusters are evident within this network. The first cluster includes prominent American research institutions and universities such as M.D. Anderson Cancer Center, Harvard University, the University of Manchester, and the University of North Carolina (Fig. 2b). The second cluster primarily comprises Chinese universities, with the Chinese Academy of Sciences at its core, Fudan University, Zhejiang University, and Nanjing Medical University. Both clusters exhibit strong intra-group collaboration and robust inter-group connections. This reflects the concentration of research forces in this field between China and the United States, and the formation of cross-cluster academic interaction. However, China's foreign cooperation is still mainly with the United States, and its ability to independently build global networks needs to be strengthened (Fig. 2b). MD Anderson Cancer Center collaborates extensively with 64 partner institutions worldwide, spanning countries such as China, the Czech Republic, Belgium, and Iran. These partnerships encompass a variety of sectors, including research institutes, universities, and pharmaceutical companies (refer to Supplementary Table S2). Similarly, the Dana-Farber Cancer Institute has 46 partners, while the National Cancer Institute has 40. These data underscore the prevalence of collaborative efforts among top-tier global research institutions. It also reflects the breadth of international cooperation among top American institutions. This is echoed by its high centrality, further consolidating its dominant position in the worldwide network (refer to Supplementary Table S2).

### 3.4 Journals and co-cited journals

We quantitatively analyzed relevant literature to understand the publication trends and journals focusing on paclitaxel resistance in breast cancer. Our study encompassed 736 journals involving diverse fields, such as oncology, pharmacology, pharmacy, biochemistry, and molecular biology, that published relevant studies. Notably, *Cancer Research* leads with 96 published articles, followed closely by *Clinical Cancer Research* (91), *Breast Cancer Research and Treatment* (75), *Molecular Cancer Therapeutics* (65), and *Oncotarget* (64). *PLoS One*, *International Journal of Molecular Sciences*, *Cancers*, *Cancer Chemotherapy and Pharmacology*, and *Anticancer Research* ranked sixth to tenth with 59, 56, 56, 53, and 51 publications, respectively. These journals are the main places where the research results of paclitaxel resistance in breast cancer are published. It also reflects the centrality of this research topic in cancer research (Table 3). In terms of impact factor (IF) in 2023, *Cancer Research* tops the list with an IF of 12.4 (JCR Q1), followed by *Clinical Cancer Research* with an IF of 10.2 (JCR Q1) (Table 3). It is also worth noting that seven of the top-ranked journals are from the United States. The dominant position of the United States in academic communication in this field was consolidated. At the same time, the journals *International Journal of Molecular Sciences* and *Cancers* are issued by Switzerland, while Greece issues *Anticancer Research* (Table 3). Overall, research on paclitaxel resistance in breast cancer predominantly appears in high-impact cancer-related journals. These journals play an important role in scholarly recognition and disseminating research in the field.

**Table 3** The top 10 journals ranked by publication number and co-citations

| Publication | | | | | | Co-citations | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Rank** | **Journal** | **Count** | **IF (2023)** | **JCR (2023)** | **Issued Country** | **Journal** | **Count** | **IF (2023)** | **JCR (2023)** | **Issued country** |
| 1 | Cancer Research | 96 | 12.4 | Q1 | USA | Cancer Research | 9057 | 12.4 | Q1 | USA |
| 2 | Clinical Cancer Research | 91 | 10.2 | Q1 | USA | Journal of Clinical Oncology | 8454 | 41.1 | Q1 | USA |
| 3 | Breast Cancer Research and Treatment | 75 | 2.8 | Q2 | USA | Clinical Cancer Research | 5430 | 10.2 | Q1 | USA |
| 4 | Molecular Cancer Therapeutics | 65 | 5.3 | Q1 | USA | Journal of Biological Chemistry | 3546 | 3.9 | Q2 | USA |
| 5 | Oncotarget | 64 | / | / | USA | Proceedings of the National Academy of Sciences of the United States of America | 3426 | 9.2 | Q1 | USA |
| 6 | PLoS One | 59 | 2.8 | Q1 | USA | Oncogene | 3328 | 6.8 | Q1 | ENGLAND |
| 7 | International Journal of Molecular Sciences | 56 | 4.3 | Q1 | SWITZERLAND | Nature | 2596 | 49.8 | Q1 | ENGLAN |
| 8 | Cancers | 56 | 4.0 | Q1 | SWITZERLAND | Annals of Oncology | 2521 | 56.2 | Q1 | NETHERLANDS |
| 9 | Cancer Chemotherapy and Pharmacology | 53 | 2.6 | Q3 | USA | British Journal of Cancer | 2440 | 6.3 | Q1 | ENGLAND |
| 10 | Anticancer Research | 51 | 1.3 | Q4 | GREECE | New England Journal of Medicine | 2324 | 95.6 | Q1 | USA |

We also analyzed the co-cited journals to gain insight into key knowledge sources. Each top 10 co-cited journals had over 2,000 co-citations (Table 3). *Cancer Research* led with 9,057 co-citations, followed by the *Journal of Clinical Oncology* (8454) and *Clinical Cancer Research* (5430). Journals ranking fourth through tenth included the *Journal of Biological Chemistry*, *Proceedings of the National Academy of Sciences of the United States of America* (PNAS), *Oncogene*, *Nature*, *Annals of Oncology*, *British Journal of Cancer*, and *New England Journal of Medicine*. Despite not publishing articles on paclitaxel resistance in breast cancer, *Nature* and the *New England Journal of Medicine* achieved high co-citation counts of 2596 and 2324 (Table 3), underscoring their importance as reference sources for advancing related research. It also shows that the study relies on the standard theories and methods of basic medicine and clinical oncology, and its knowledge source has an interdisciplinary breadth.

Although the Journal of Clinical Oncology has fewer articles (40 publications), it boasts an impressive total co-citation count of 8,454 (Table 3). Similarly, each *Journal of Biological Chemistry*, *PANS*, and *Oncogene* has garnered over 3,000 co-citations, surpassing even renowned journals like *Nature* and the *New England Journal of Medicine* (Table 3). These findings indicate that research into paclitaxel resistance in breast cancer heavily relies on reputable scholarly journals as its knowledge base. Notably, with an impact factor of 3.9 in 2023, the Journal of Biological Chemistry achieved a remarkable co-citation count of 3,546, ranking fourth. These findings underscore its significant role as a source of knowledge in this field.

Supplementary Fig.S2 presents a co-occurrence network of the top 5% most-cited journals. By integrating insights from Table 3 and Supplementary Fig.S2, it is evident that highly cited journals such as *Cancer Research*, *Journal of Clinical Oncology*, *Clinical Cancer Research*, and *Annals of Oncology* have maintained citation spans ranging from 20 to 30 years. Notably, *Oncotarget*, launched in 2010 as an oncology-focused, peer-reviewed open-access journal, stands out despite being delisted from the Science Citation Index (SCI) in 2018. It consistently ranks high with 64 publications and 2737 citations, receiving frequent citations up to 2024 (Table 3), which suggests that researchers' reliance on knowledge sources extends beyond SCI inclusion.

Additionally, several journals like *Biomaterials*, *Molecular Pharmaceutics*, *the Journal of the American Academy of Pediatrics*, *International Journal of Molecular Sciences*, and *International Journal of Pharmaceutics* are noteworthy. Although these journals entered the field more recently, they have achieved high publication counts. It indicates that research may be expanded to drug delivery, nanotechnology, and other cross-disciplinary areas. They may become new knowledge transmission and source nodes (Table 3).

### 3.5 Co-cited references and citation bursts

Co-cited references refer to documents jointly cited by two or more publications. Analyzing co-cited references within a literature collection over a specific period allows researchers to gain insights into the origins and advancement in relevant research fields. Employing VOSviewer software, we conducted a co-citation analysis on 3,668 documents. The top 10 most frequently co-cited references are listed in Supplementary Table S3. The most frequently co-cited reference is "*Multidrug resistance in cancer: role of ATP-dependent transporters*" by Michael M Gottesman et al. (2002) [26], with a co-citation count of 194 (Supplementary Tab S3). This article systematically reviewed the mechanism of multidrug resistance in cancer and has played a fundamental role in research. It is followed by "*Microtubules as a target for anticancer drugs*" by Mary Ann Jordan et al. (2004)[3] and "*Cancer Statistics,2021* "by Siegel RL et al. (2021)[1] (Supplementary Table S3). what these references were frequantly co-cited suggest that exploration of basic mechanism and clinical data support are key pillars in the development of this field.

Additionally, several articles focusing on mechanisms of Taxol resistance related to microtubules are also highly co-cited, including "*Mechanisms of Taxol resistance related to microtubules*" by George A Orr et al. (2003)[27], "*Taxanes, microtubules and chemoresistant breast cancer*" by McGrogan BT et al. (2008)[28], and"*Microtubules and resistance to tubulin-binding agents*" by Kavallaris M et al. (2010)[29] (Supplementary Table S3). Research into microtubule dysfunction and associated changes in protein expression represented a key node in this field's progression and knowledge inheritance. Furthermore, a clinical trial conducted by D J Slamon et al. [7], titled "*Use of chemotherapy plus a monoclonal antibody against HER2 for metastatic breast cancer that overexpresses*

*HER2*" has reveived 117 co-citations, with an average of 5.09 citations per year (Supplementary Table S3), highlighting its significance in understanding paclitaxel resistant in breast cancer.

Citation bursts occur when literature receives a significantly higher frequency of citations than the average over two years. By visualizing citation bursts on a timeline, researchers can intuitively understand the shifts in research hotspots. We used CiteSpace to analyze the citation burst strength of 3,668 documents. The top 25 strongest bursts are shown in Fig. 3a and Supplementary Table S4, ordered by the year each burst began. These data reveal the developmental trajectory of paclitaxel resistance research. Specifically, Holmes et al.'s 1991 study[30], which experienced a citation burst from 1993 to 1996 (Fig. 3a), emphasized the importance of investigating optimal drug combination sequences and identifying dose-limiting toxicities, particularly cardiac toxicity associated with taxol. It also highlighted that the optimal dose and sequence of combination therapy require further investigation (Supplementary Table S4). Subsequently, Wilson's 1994 study[31] on the timing and optimal dosage of paclitaxel infusion received significant citations (Supplementary Table S4), providing a foundation for clinical dose individualization and experiencing a citation burst from 1995 to 1999. Collectively, from 1991 to 1999, medical scientists primarily focused on determining optimal dosing methods, administration techniques, and the clinical efficacy of paclitaxel (Supplementary Table S4). These studies overcame key clinical application barriers and established paclitaxel's role in the treatment of breast cancer. However, an in-depth understanding of paclitaxel's molecular targets and the potential mechanisms underlying therapeutic failure remains limited.

As the 2000 s approached, research on the mechanisms of paclitaxel resistance and clinical trials to overcome this resistance experienced a notable surge in citations (Fig. 3a and Supplementary Table S4). Between 2000 and 2008, key studies include "Microtubules as a target for anticancer drugs" by Mary Ann Jordan et al. (2004)[3], "Class III beta-tubulin overexpression is a prominent mechanism of paclitaxel resistance in ovarian cancer patients" by Mozzetti et al. (2005)[32], and "Microtubule-associated protein tau: a marker of paclitaxel sensitivity in breast cancer" by Rouzer R et al. (2005)[33], all of which focused on elucidating the mechanism underlying paclitaxel resistance and were subsequently cited extensively (Fig. 3a and Supplementary Table S4). Furthermore, clinical trials evaluating ixabepilone for metastatic breast cancer, both as a monotherapy and in combination with capecitabine or bevacizumab, also garnered increased citations, suggesting that researchers were actively investigating novel therapeutic approaches and combination strategies during this period (Fig. 3a and Supplementary Table S4).

Since 2008, there has been a significant rise in citations for review articles. McGrogan BT (2008)[28], Kavallaris M (2010)[29] et al. systematically integrated the mechanisms of microtubule-related drug resistance and put forward the view of "multi-mechanism cooperation leading to drug resistance", which provides a basis for multi-target intervention (Fig. 3a and Supplementary Table S4). Studies on drug resistance in special subtypes of triple-negative breast cancer (TNBC) (Schmid, 2018)[34] have shown that paclitaxel combined with immunotherapy can prolong the survival of TNBC patients, and immunotherapy should be included in the strategy to overcome drug resistance (Fig. 3a and Supplementary Table S4). Nanocarriers have partially reversed drug resistance by improving drug delivery efficiency and reducing toxicity (Abu Samaan, 2019)[35], laying

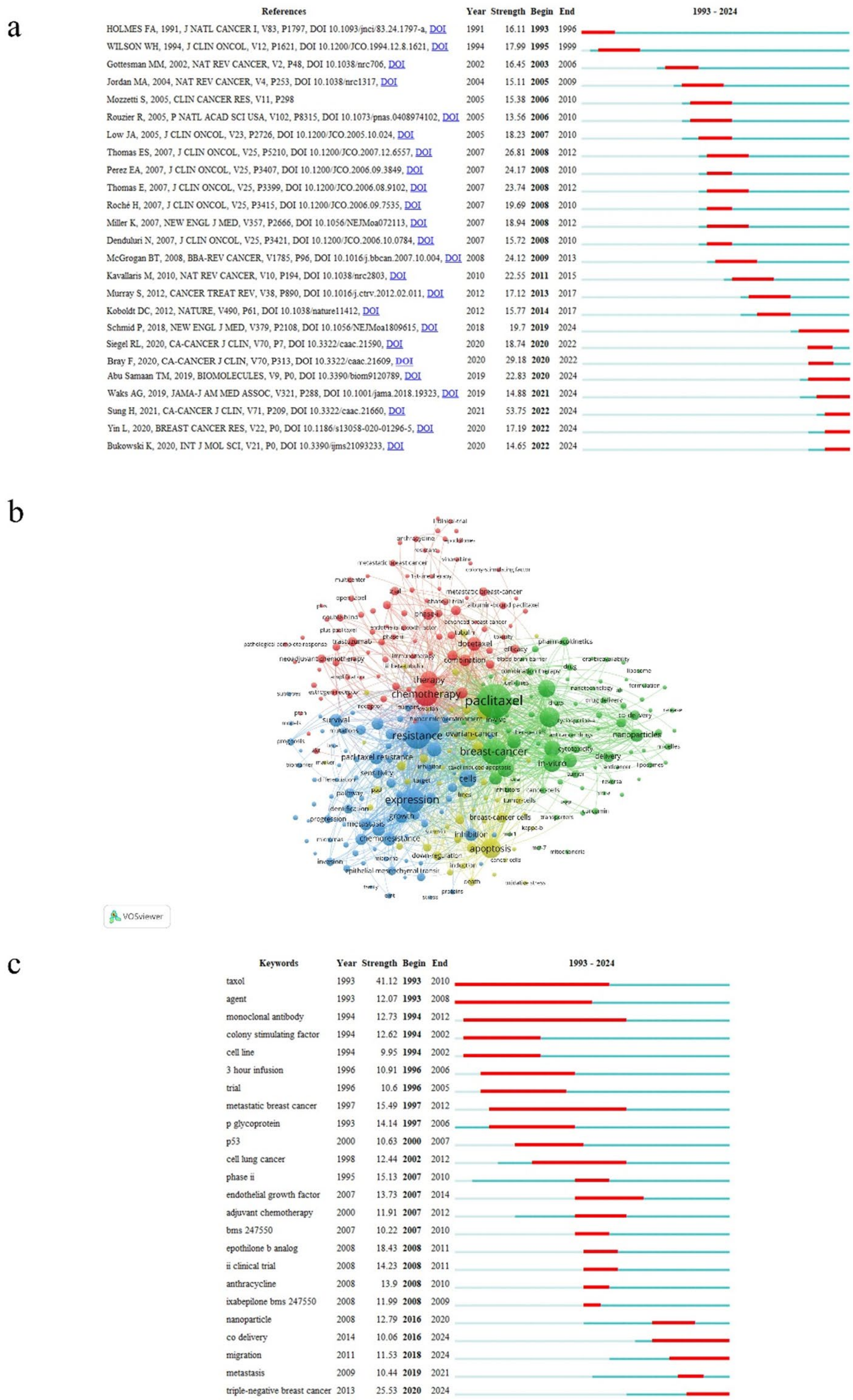


**Fig. 3** Citation bursts and clusters. **a** Top 25 references ranked by the strongest citation bursts. (The blue line denotes the observation period of the literature from its publication year, while the red line denotes the duration of the citation burst). **b** The clusters of keywords. (A cluster analysis of keywords whose occurrence frequency was more than 20 times was performed using VOSviewer software. In this visualization, colors denote clusters, dot size indicates occurrence frequencies, and connecting lines signify the co-occurrence relationship. **c** Top 25 keywords ranked by the strongest citation bursts

the foundation for targeted delivery (Fig. 3a and Supplementary Table S4). During this period, research on drug resistance mechanisms has expanded from a single molecule to a network of pathways. Nanotechnology, immunotherapy, and other emerging technologies provide new ideas for reversing drug resistance.

The evolution of citation patterns over the past three decades reflects advancements in this field. Research on paclitaxel-resistant breast cancer has advanced from initial clinical experiments to mechanistic investigation and from identifying drug resistance mechanisms to exploring strategies to overcome them. Each step is based on the accumulation and breakthrough of previous problems. However, some core problems, such as drug resistance heterogeneity, lack of predictive markers, and insufficient targeting of nano-delivery systems, need to be solved by interdisciplinary collaboration.

### 3.6 Research directions, hot spots, and trends

Keywords accurately capture and summarise an article's main topics and essence. Generally, articles on similar subjects will share one or more keywords. By aggregating keywords from all articles within a research domain, we can uncover the current research focuses and emerging trends in specific areas [36]. The research directions, hot spots, and trends of paclitaxel resistance in breast cancer were analyzed through keywords co-occurrence and cluster analysis. A total of 10,851 keywords were presented in studies on paclitaxel resistance in breast cancer. 'Breast cancer' (n = 1,639), 'paclitaxel' (n = 1,558), and 'resistance' (n = 873) were the core keywords with significantly higher frequency than other words. It directly anchors the core objects, intervention drugs, and core questions of the study, and is the basic framework for research in this field. Other high-frequency keywords included 'expression' (n = 773), 'multidrug-resistance' (n = 732), 'chemotherapy' (n = 723), 'drug resistance' (n = 613), 'apoptosis' (n = 516), 'in-vitro'(n = 403), and 'doxorubicin' (n = 394). It further reflects expanding research on the drug resistance phenomenon, research methods, and clinical applications. It reflects ongoing interest in resistance mechanisms and clinical correlates (Supplementary Table S5).

To help researchers better understand the knowledge landscape and identify research hotspots and trends, we used VOSviewer software to perform a cluster analysis of the 313 keywords whose occurrence frequency was more than 20 times. As shown in Fig. 3b, the 313 keywords were categorized into four clusters. The red clusters include 'chemotherapy', 'docetaxel', 'combination', 'phase-ii', and other keywords related to treatment methods, reflecting the research direction of clinical treatment strategies. The green clustering includes keywords such as 'delivery', 'drug', 'nanoparticles', 'release', 'liposome', and 'in-vitro', which are mainly related to drug delivery and reflect the application and exploration of technical means in improving drug resistance. The blue clusters include 'expression', 'activation', 'pathway', 'survival', 'sensitivity', 'target', and other keywords mainly related to mechanisms. It is the basic research field that analyzes the nature of drug resistance. The yellow clustering includes keywords such as 'apoptosis', 'induction', 'death', 'survivin', et al., which are mainly related to the cellular response and reflect studies on changes in the physiological state of cells during drug resistance (Fig. 3b). These four clusters represent four major branches in paclitaxel-resistant breast cancer research. This clustering structure indicates that the research in this field has formed a multi-dimensional and structured system. The branches are both independently focused and interrelated.

Keyword burst detection analysis reveals that from 1993 to 2006, frequently cited keywords included 'taxol', 'monoclonal antibody', 'metastatic breast cancer', and 'p glycoprotein'. It reflects the early exploration of drugs, classical drug resistance proteins, and basic targeted therapy. It is in the stage of preliminary discovery of drug resistance and initial exploration of its mechanism (Fig. 3c). From 2007 to 2015, the focus shifted to ixabepilone and adjuvant chemotherapy. It reflects the expansion of research to the combination of chemotherapy drugs and adjuvant therapy strategies, and pays more attention to optimizing clinical treatment regimens. Since 2016, emerging keywords such as 'nanoparticle', 'co delivery', 'migration', 'metastasis', 'triple-negative breast cancer' (TNBC), and 'tumor microenvironment' have gained prominence. Notably, while terms like 'co delivery', 'migration', 'TNBC', and 'tumor microenvironment', appeared over a decade ago, their citation burst only occurred after 2016 and continues to this day (Fig. 3c). This trend indicates that research has entered a new stage of technology-driven, precision and micro-environmental regulation. It reflects the trend from "mechanism exploration" to "technology transformation" and "subtype-specific research", and is deeply integrated with frontier fields such as tumor metastasis and microenvironment.

## 4 Discussions

Paclitaxel resistance remains a significant challenge in the treatment of breast cancer and has been the focus of extensive research for over three decades. Despite this body of work, a comprehensive, global overview of the development and evolution of this research field is still lacking. The present study addresses this critical gap by presenting a 31-year-long bibliometric analysis, which identified key global contributors and their collaborative networks, determined influential professional journals in this field, summarized paclitaxel resistance mechanisms, traced the evolution of central research questions, and highlighted promising future research directions. It provides a spatiotemporal geographic landscape of paclitaxel resistance in breast cancer for the first time, serving as a valuable reference for researchers in this field.

Our findings reveal a distinct pattern of knowledge diffusion and concentration. Leadership in high-impact research is predominantly held by major institutions in North America, Europe, and East Asia, with the United States at the forefront. Significantly, this leadership extends beyond mere output volume and is closely tied to extensive and highly central international collaboration networks, exemplified by institutions such as Harvard University, the Dana-Farber Cancer Institute, the National Cancer Institute, and MD Anderson Cancer Center. This observation is consistent with theories of innovation diffusion, which posit that dense collaboration networks serve as essential channels for exchanging complex knowledge, expediting scientific discovery, and shaping research standards [37]. The prominence of these collaborative hubs highlights the growing imperative for globally integrated, interdisciplinary approaches to address complex global challenges such as drug resistance, surpassing the limitations of localized or institutional efforts previously reported in the literature.

By analysis of keyword clustering and citation burst, we identify a clear evolutionary trend in the field: from the initial clinical application of paclitaxel, to in-depth investigations into the complex mechanisms of resistance, and finally to integrating this knowledge to develop tailored therapeutic strategies for distinct breast cancer subtypes. The shift from 'trial-and-error' combination regimens to mechanism-informed precision therapy mirrors Kuhn's paradigm shift model [38] of scientific revolution, where resistance mechanisms act as 'anomalies' dismantling the initial paradigm of empirical paclitaxel optimization. Significantly, emerging technologies are increasingly shaping the field. Technological innovation is a pivotal driving force, bridging the gap between mechanistic research and clinical application. The rise of nanotechnology, such as Abraxane [39], Lipusu [40], and numerous novel nanocarriers under development [41], directly addresses pharmacokinetic limitations of paclitaxel, enhancing targeting and overcoming resistance mechanisms like efflux pumps [42]. These advancements are pivotal in translating laboratory discoveries into improved patient clinical outcomes.

In summary, our study covers 31 years of literature data from 1993 to 2024, offering a multidimensional perspective on the evolution and developmental trajectory of research on paclitaxel resistance in breast cancer. Prior studies have largely concentrated on narrow aspects—such as specific molecules or signaling pathways linked to drug resistance, individual clinical trials, or narrative reviews—without providing a comprehensive overview of the field's spatiotemporal development. In contrast, our study is the first to employ bibliometric analysis to address this critical gap, mapping the spatiotemporal distribution of research hotspots and patterns of knowledge diffusion in this domain. This novel analytical approach offers researchers a macro-level and holistic understanding of the field's progression and current state.

Nevertheless, certain limitations should be acknowledged. First, our analysis was confined to English-language publications in the WOSCC database, which may introduce language biases. Second, studies published after December 31, 2024, were excluded due to citation lag, which may have resulted in an incomplete dataset. Future studies must update the dataset every three to five years to track the field development dynamically. Lastly, data extraction relies on bibliometric tools incorporating machine learning and natural language processing techniques, which, while advanced, can still have algorithmic biases [21, 43].

## 5 Conclusions

This study presents a 31-year bibliometric analysis of research on paclitaxel resistance in breast cancer, identifying the key factors driving the development of the field: global collaboration networks, the thematic evolution from drug optimization to precision medicine, and transformative technologies such as nanodelivery systems and liquid biopsy. It also highlights a strategic direction for future research, including strengthening international partnerships, deepening understanding of resistance mechanisms, and applying engineered therapeutics for targeted breast cancer treatments.

## Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1007/s12672-025-03420-3.

Supplementary material 1.

Supplementary material 2.

### Author contributions

Y.L. and Y.M. designed the study. Y.M., C.Z., H.F., and P.L. collected, processed, and analyzed data. Y.L. and T.H. supervised the research. Y.M. and Y.L. wrote the manuscript. Y.M., C.Z., P.L., H.F., W.W., S.P., T.H., and Y.L. discussed and revised the manuscript.

**Funding**
This work was supported by the Sichuan Science and Technology Program Grant 2025YFHZ0154 to Yan Lin, 2024YFFK0346 and 2022YFS0623 to Tao He, and Luzhou Science and Technology Program 2022-JYJ-133 to Y.L.

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**Data availability**
Data is provided within the manuscript or supplementary information files.The data generated in the present study may be requested from the corresponding author.

## Declarations

**Consent for publication**
All authors know and agree to the paper's content and their being listed as co-authors.

**Competing interests**
The authors declare no competing interests.
Ethics and consent to participate declarations.
Not applicable.

Received: 14 May 2025 / Accepted: 11 August 2025

Published online: 24 August 2025

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